

~~DT 4/30/18 9:50am sent to Janet for approval~~



Southern California Edison
CS Re-Platform Project

CIM-CS-05-Handle Request-Process Field Work
Request

Customer Interaction Management

Business Process Design

Version 1.9

Table of Contents

1	Business Process Overview.....	5
1.3	Current Pain Points for Southern California Edison	5
1.2	Responsible Parties.....	5
1.3	Document Control.....	6
1.4	Terminology	6
2	Business Process Design	8
2.1	Manage Field Work Request Details	8
2.1.2	Work Order Creation for SCE Customers	10
2.1.3	Work Order Creation for non-SCE Customers	10
2.1.4	Work Order Completion	12
2.2	Process / Performance Improvement	12
2.3	Prerequisites and Inputs	12
2.4	General Assumptions	13
2.5	Business Process Hierarchy	13
2.6	Process Flows	13
2.6.1	Work Order Processing	14
2.7	3 rd Party and SCE Legacy System Integrations.....	16
2.8	Data Elements	16
2.8.1	Service Order Data Elements	16
	Please refer to the BPD DF-WM-02 - Handle Field Orders - Manage Field Orders for detail data elements.....	16
2.8.2	Source Systems and Data Volumes	16
2.9	Cloud for Customer (C4C) Integration	16
2.9.1	Processes	16
2.9.2	C4C Views	16
2.9.3	Activity	17 16
2.10	Exception Management Requirements.....	17
2.11	Process Risks and Mitigation.....	17
2.12	Key Performance Indicators	17
2.12.1	Key Metric Definition.....	17
3	Controls, Regulatory Rules, and Security	18
3.1	Security Considerations.....	19
3.2	Custom Program/Transaction Codes.....	20
3.3	Custom Table/Transaction Codes	20
3.4	Data Criticality and Cybersecurity Baseline Risks	20
4	Technical Considerations	22
4.1	RICEF-W Assessment.....	22
4.1.1	Reports	22
4.1.2	Interfaces	22
4.1.3	Conversion	22
4.1.4	Extension/Enhancements	22
4.1.5	Forms.....	22
4.1.6	Workflows	23
4.2	IDEX Considerations	23

4.3	Production System Cut-over Considerations	23
4.4	Other Considerations.....	23
5	Change Management	24
5.1	Change Impacts.....	24
6	Requirements.....	28

This page intentionally left blank

1 Business Process Overview

SAP work order is used as a document to log a customer's request which will then be used to communicate with the appropriate business unit for a field visit. It provides details required by field workers such as requested date, requested time, note containing access instructions, tasks, customer, location, etc.

There are many scenarios that will require an order to be sent to the field and they are differentiated by category in order for right type of skilled resource assignment to carry out the task(s).

Example of potential scenarios may include:

- Submitting and tracking a request for field work
 - Disconnection, Inspection, Installation
- Reassigning field work for completion
- Automatically generating field work order based on customer web interaction
- Canceling a request for field work

This document captures the process for the creation of Service Orders (SO) and their related Service Notification (SN) for field work order request via Hybris C4C to integrate with the Work Management (WM) processes for SCE.

1.3 Current Pain Points for Southern California Edison

- Prioritization can be a problem. Based on manpower some orders are not assessed by field as being a priority.
- Info / notes field for the BWR request is limited (ex. 70 CHARS)
- Difficult for agents to see the notes which are made in the field
- Field reps cannot see the additional notes (they have to call the dispatcher who will look up the customer / BWR and read them the notes over the phone)
- Field reps will complete the order without actually completing the work and without notes.
This results in billing not understanding why the work was not completed and the creation of a new Order which now reverts to the back of the schedule rather than being treated as a priority or re-issue.

1.2 Responsible Parties

Contact	Contact Person
Southern California Edison Business Process Owner	Janet Covington
Southern California Edison Lead	Christine Evans
HCL Functional Lead	Adrianus Halim
Solution Architect	Pedro Costa
Front Office or Back Office Lead	Christine Evans

Contact	Contact Person
Subject Matter Experts	Christine Evans Cari Young Ernesto Valle Pedro Costa
Key User Groups	Customer Contact Center (CCC), BCD

1.3 Document Control

Version	Description of Change	Date	Author
1.0	Creation of Initial Version	01/02/2018	Satyendra Singh
1.1	First review from HCL CIM Lead	02/07/2018	Adrianus Halim
1.2	Updated Visio	02/14/2018	Jessica Marsillo
1.3	Updates to RICEFW Table	02/23/2018	Jessica Marsillo
1.4	Inputs from D&F Team	03/01/2018	Adrianus Halim
1.5	Update to Visio	03/02/2018	Jessica Marsillo
1.6	Updates from the review	03/02/2018	Adrianus Halim
1.7	Updates from the D&F Team	03/06/2018	Adrianus Halim
1.8	Updates after meeting with PnP	03/14/2018	Adrianus Halim
1.9	Updates from SME review	04/27/18	Adrianus Halim

1.4 Terminology

Term/Acronym	Definition and SAP Equivalent (if applicable)
APSE	Accelerated (or Axon) Project Support Environment
BDEX	SAP add-on functionality to support the processing of BPEM Cases
Billing Cycle	Is a group of contracts to be billed together via date-driven or regional criteria, it controls the billing cycle and is linked to the Meter Reading Unit. SAP equivalent term is the Portion.
Billing Document	A document that outlines the charges incurred for energy consumption for a specific contract. During the process of Billing, a billing document is generated for each contract on a contract account.
BMD	Business Master Data. The customer information which includes personal details, payment terms, dunning levels, billing rates and rate categories. The business master data objects are Business Partner, Business Agreement/Contract Account and Contract.
BPD	Business Process Design (or Document)
BPEM	Business Process Exception Management

Term/Acronym	Definition and SAP Equivalent (if applicable)
BWR	Basic Work Request. SAP equivalent is BPEM.
CA	Contract Account
Channel	Data object in SAP that holds the interval data. Intervals can be kwh, price, factor, etc. SAP equivalent is Profile.
Clarification	A list of payments that could not be processed and require additional work in order to be posted.
Connection Object	Link between installation and postal regional structure. Usually a building, but can also be a piece of property or other facility, such as a fountain or construction site. The connection object corresponds to the functional location in plant maintenance (PM).
Contract	Standard SAP object that is used to link SAP Business Master Data and Technical Master Data through the move-in process
Customer	Any entity with a relationship with SCE. This term is usually used in reference to customers (both individuals and organizations) This can include employees and third party suppliers (banks and payment partners). SAP Equivalent is Business Partner (BP)
EAM	SAP Enterprise Asset Management
Hybris C4C	SAP Cloud for Customer Solution
IMG	SAP Implementation Guide – where configuration is performed
Level Pay Plans	SAP equivalent term is Budget Billing. Budget Billing allows customers to pay regular equal amounts over a yearly period.
OA	Owner Allocation
Premise	Represents one of the SAP Technical Master Data object that is supplied with energy, for example, an apartment or factory. A premise can have several utility installations. The premise is allocated to a connection object and the address of that connection object
Property	Standard SAP object that is primarily used for arranging agreements with the owner concerning automatic move-in. It also stores an owner history for installations that are part of owner allocation process
S/4HANA	SAP HANA 4 Business Suite
SCE	Southern California Edison
Summary Billing	Standard SAP functionality that enables the invoicing of multiple Contract Accounts under one Parent Contract Account (Consolidated Summary Billing). SAP equivalent is Collective Billing.
TCODE	Transaction Code Description
TMD	SAP Technical Master Data e.g. Connection Object, Premise and Device Location

Additional terms may be found in APSE [General Document # 32 CSRP Glossary](#)

2 Business Process Design

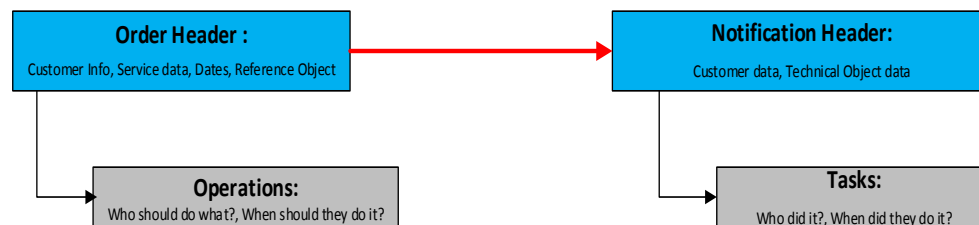
The SO generates a request for a field resource. There are variety of scenarios that require an order to be sent to the field. For each SO created in the system, a specific order type is selected from a drop-down list prior to the creation itself. The selected type is important because based on this, a field resource with the right skills for that work will be assigned. The SO can also be automatically created by the system in response to an event (ex. Customer moves into a premise with a legacy meter requiring a field visit). The enhancement E-CIM-CS-05 Order Creation has been identified to provide necessary fields that are needed when creating or changing work order.

This document details the process to manage the creation and initiation of SO via Hybris C4C. The handling of the order will covered in DF-WM-02 - Handle Field Orders – Manage Field Orders.

2.1 Manage Field Work Request Details

The SO is used to document the work which needs to be performed in the field. This will include the details of the type of work to be performed, the technician assigned to the work, the location and/or customer for whom the work is to be performed, who created the order etc. The SN is created automatically in the background based on the order, via SAP configuration. This is where it maintains history of work progress and the tasks used to capture work results, including the existing completion codes. Items, tasks and activities can be entered in notifications. The SN is closed automatically when orders are technically completed.

Below shows the main SAP components when using the creation and management of SO.



The following are the description of the items in the above picture:

Order Header: Describes the job, order type, PM activity type representing dispatch code, Date, Reference Object (i.e. the meter, device location, connection object), Address Information, Customer information, Priority, Status.

Operations: Detailed steps of the job to be carried out. The operation description is copied from the service order header short text field.

Notification Header: Describes Customer data, Technical object data and execution plan. Used to maintain history of work completion, used on mobile device to update the status of order (Completion Code)

Tasks: Describes what job has been done and status. Tasks will have task codes and each code is used as completion code

When creating a new SO via Hybris C4C, the users normally need to confirm the appropriate objects (ex. Business Partner (BP) and / or Premise).With reference to Figure 1 below, they will then select the appropriate order type which will be a required field and drive the assignation of the appropriate field staff to the order based on the order type like maintenance, installation, or even investigation order. There are also orders that can be manually created as a result of customer program enrollment request such as API and SDP RF Devices, Customer DR

Enabling Technology, Home Area Network, etc. The initiation point will be covered in the BPD [PP-CS-03 - Fulfill Customer Request](#).

The user will be able to select the service order type from a drop down (BPD [DF-WM-02 - Handle Field Orders – Manage Field Orders](#)). The user will be able to select an appointment date/time (e.g., 4 hour window based on field resource availability) so that the field person can meet the customer or other party at the site. This can be due to access issues and additional information that field person may need from the customer while onsite. (BPD [DF-WM-02 - Handle Field Orders – Manage Field Orders](#))

They will then complete the order details (ex. date / time, notes and access related instructions, etc.) and submit the order. The enhancement E-CIM-CS-05 Order Creation has been identified to provide necessary fields that are needed when creating a new SO.

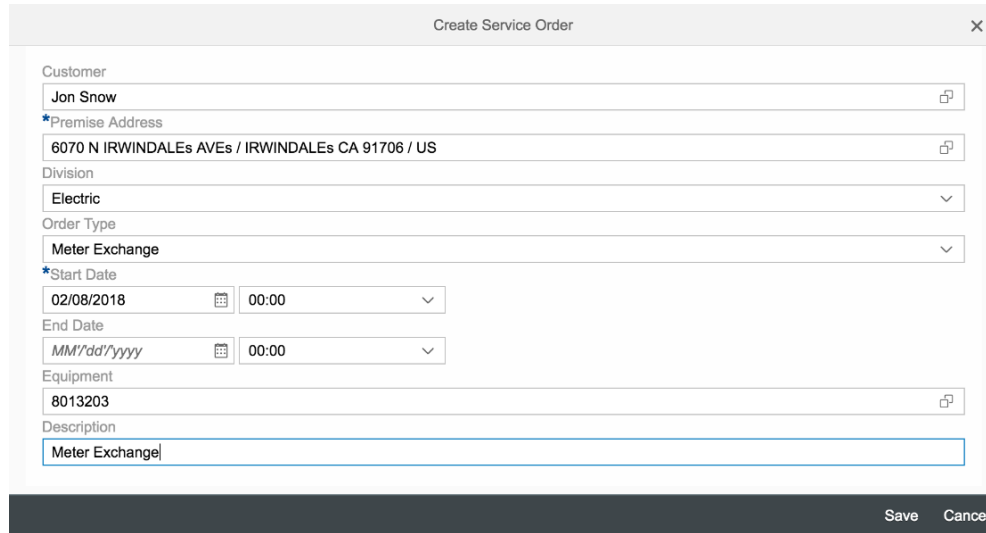
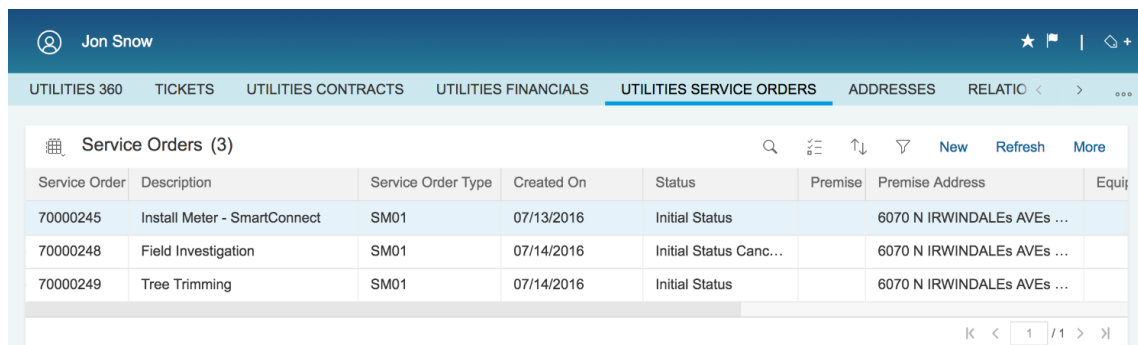


Figure 1 – Create a New Service Order

Once a new SO has been created, the order will be handled based on what is required in order to get the assigned tasks done. This will be covered in BPD [DF-WM-02 - Handle Field Orders – Manage Field Orders](#).

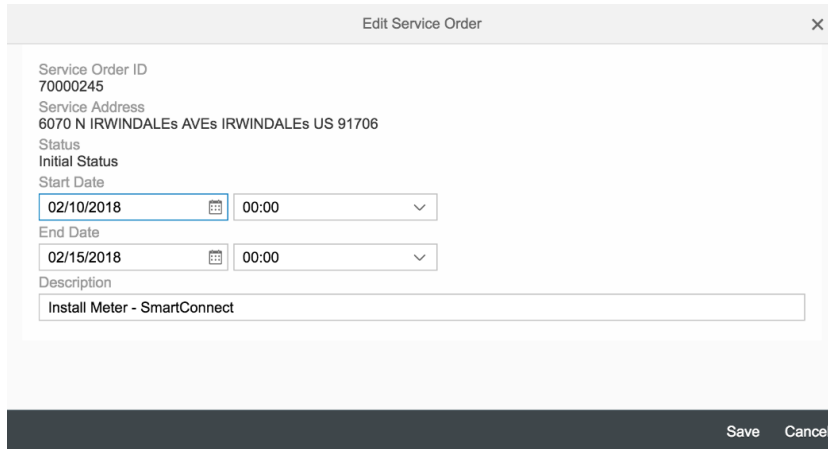
Orders that are linked to a customer are viewable in Hybris C4C with creation dates, status (open vs. completed, etc.). The order status will be used to as a reference by the users to determine where the order is at in its lifecycle (ex. in field, onsite, etc.).



Service Order	Description	Service Order Type	Created On	Status	Premise	Premise Address	Equip
70000245	Install Meter - SmartConnect	SM01	07/13/2016	Initial Status		6070 N IRWINDALES AVES ...	
70000248	Field Investigation	SM01	07/14/2016	Initial Status Canc...		6070 N IRWINDALES AVES ...	
70000249	Tree Trimming	SM01	07/14/2016	Initial Status		6070 N IRWINDALES AVES ...	

Figure 2 – Service Order List

The authorized users will also be able to perform the edit to existing order in Hybris C4C and the enhancement E-CIM-CS-05 Order Creation has been identified to provide necessary fields that are needed when changing existing SO.



Service Order ID
70000245

Service Address
6070 N IRWINDALES AVEs IRWINDALES US 91706

Status
Initial Status

Start Date
02/10/2018 00:00

End Date
02/15/2018 00:00

Description
Install Meter - SmartConnect

Save Cancel

Figure 3 – Edit Service Order

2.1.2 Work Order Creation for SCE Customers

When a customer calls SCE for service, the user creates a new SO by selecting appropriate order type (refer to figure 1) that is relevant to customer's request. A SN will be created in the background automatically. A unique number is assigned to the order by the system automatically. Confirming a BP and premise will be key criteria (by SAP Standard functionality) when creating a work request for the field personnel to carry out the task. The list of service orders can be shown in C4C showing the type of orders created with current status for the specific location. (Refer to Figure 2)

2.1.3 Work Order Creation for non-SCE Customers

There is a scenario in which either BP and/or premise cannot be found due to the fact that the caller is a person who does not have any business associated with SCE trying to inform about issue at certain location. Such caller can be categorized as non-SCE customers who reports a problem with a location that cannot be located in the system. The enhancement E-CIM-CS-05 Order Creation will be used to handle this scenario (see figure 4 for backoffice screenshot). Again, the order type selection will still be the main key component to be selected determining the type of order as starting point. The field order note can be utilized to record the additional problem reported by the caller (see figure 5 for backoffice screenshot)

Figure 4 – Non-SCE customer and Non-Standard Address

Figure 5 – Order note capturing additional information

The next step is to select a district by calling the enhancement E-DF-WM-02 - District Determination (see Figure 6 for backoffice screenshot) to determine the dispatched area where particular service is required to be carried out. In addition to this, the field PM Activity Type is also used to categorize the job code from the drop-down field. These two fields and work order routing are to be further detailed in the BPD [DF-WM-02 - Handle Field Orders – Manage Field Orders](#).

PI	Search Ter...	Search Ter...	Postl Co.	City	Name 2	Name
22			91754	Monterey Park		MONTEBELLO FIELD LOCATION
26			91773	San Dimas		COVINA FIELD LOCATION
27			91016	Monrovia	MONROVIA FIELD LOCATION	MONROVIA FIELD LOCATION
29			92750	Santa Ana	SANTA ANA FIELD LOCATION	SANTA ANA FIELD LOCATION
30			92336	Fontana	FOOTHILL FIELD LOCATION	FOOTHILL FIELD LOCATION
31			92373	Redland		REDLANDS FIELD LOCATION
32			90220	Compton		COMPTON FIELD LOCATION
33			92646	Huntington Beach		HUNTINGTON BEACH FIELD LOCATION
34			91761	Ontario		ONTARIO FIELD LOCATION
35			91361	Thousand Oaks		THOUSAND OAKS FIELD LOCATION
36			93539	Lancaster		ANTELOPE VALLEY FIELD LOCATION
39			93004	Ventura		VENTURA FIELD LOCATION
40			92378	Rimforest		ARROWHEAD FIELD LOCATION
42			90404	Santa Monica		SANTA MONICA FIELD LOCATION
43			92618	Irvine		SADDLEBACK FIELD LOCATION
44			90503	Torrance		SOUTH BAY FIELD LOCATION
46			90806	Long Beach		LONG BEACH FIELD LOCATION
47			90670	Santa Fe Springs		WHITTIER FIELD LOCATION
48			92833	Fullerton		FULLERTON FIELD LOCATION
49			93117	Goleta		SANTA BARBARA FIELD LOCATION
50			93664	Monterey Park		SHAVER LAKE FIELD LOCATION
51			93274	Tulare		TULARE SERVICE CENTER
52			93561	Tehachapi		TEHACHAPI FIELD LOCATION
53			93285	Wofford Heights		KERNVILLE FIELD LOCATION
59			91355	Valencia		VALENCIA FIELD LOCATION
61			90704	Avalon		CATALINA FIELD LOCATION
72			92311	Barstow		BARSTOW FIELD LOCATION
73			92392	Victorville		VICTORVILLE FIELD LOCATION
77			92585	Romoland		MENIFEE FIELD LOCATION
79			92234	Cathedral City		PALM SPRINGS FIELD LOCATION
85			93514	Bishop		BISHOP FIELD LOCATION
86			93555	Ridgecrest		RIDGECREST FIELD LOCATION
87			92225	Blythe		BLYTHE FIELD LOCATION
88			92595	Wildomar		WILDOMAR FIELD LOCATION

Figure 6 – Dispatched Areas

2.1.4 Work Order Completion

The C4C user will be able to view the status of the work order throughout the life cycle of the work order (BPD [DF-WM-02 - Handle Field Orders – Manage Field Orders](#)). The list of work order completion status will be addressed in D&F BPD.

If additional action is needed by a different department (e.g., CCC, RSO, etc.) to complete the service order and/or customer request, T&D will initiate a request to the required department. (BPD [DF-WM-02 - Handle Field Orders – Manage Field Orders](#)).

2.2 Process / Performance Improvement

- Improved reporting data capabilities.
- The work orders will be stored in S/4HANA and there are useful standard SAP reports that can be used for order lifecycle improvement, order tracking, and customer service satisfaction.
- Users can see work orders related to a customer in the system (ie. Work order type, status, etc).
- Work order history log to provide better information when there is a need to perform investigation.

2.3 Prerequisites and Inputs

- Business Partner
- Premise (if required)
- Order Type

2.4 General Assumptions

- Click will be served as scheduler to assign the skilled personnel and work duration based on order type after the order has been created
- There will be an interface between S4/Hana and Click to create the request in Click to be scheduled. (Refer to BPD [DF-WM-02 - Handle Field Orders – Manage Field Orders](#))
- There will be an inbound interface from Click to S/4HANA to update and complete the service order with status and comments (Refer to BPD [DF-WM-02 - Handle Field Orders – Manage Field Orders](#))

2.5 Business Process Hierarchy

This Business Process Hierarchy provides a listing of all scenarios/business processes, sub-processes, and SAP Transaction codes identified to support this process.

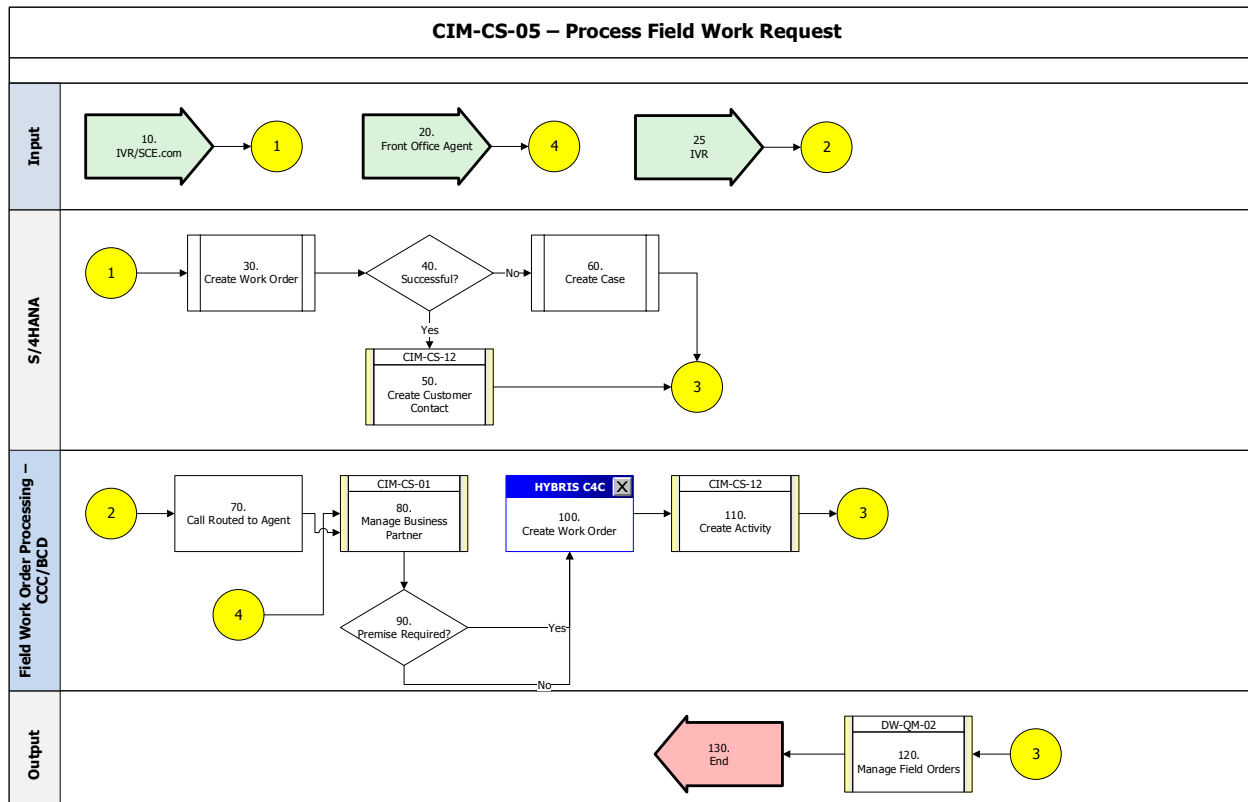
Level 1	Level 2	Level 3	Level 4
Back Office (CCC, GC Services/BCD/RSO)	Process Field Work Request	Display Service Order	IW33
Front Office (CCC, GC Services/BCD)	Process Field Work Request	Create Service Order Change Service Order Display Service Order	Front Office/C4C IW31 IW32 IW33
CCC, GC Services/BCD	View Field Work Order	Display Service Order	IW33

2.6 Process Flows

This BPD will cover 1 process, including Work Order Processing. Each subsection where the process is detailed will have a process flow diagram for that process and a table detailing the steps that are taken within that process including processes handled outside of SAP.

2.6.1 Work Order Processing

Process Flow Diagram



Process Flow Steps

Step	Description of Step	T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
10	Start Customer starts Work Order request at IVR/SCE.com. Proceed to Step 30.	N/A	IVR/SCE.com	
20	Start Front Office Agent receives request from the customer. Proceed to Step 80	N/A	Field Work Order Processing – CCC/BCD	
25	Start Customer request via IVR receives request from the customer. Proceed to Step 70	N/A	Field Work Order Processing – CCC/BCD	

Step	Description of Step	T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
30	Create Work Order	S/4HANA IW31	S/4HANA	
40	Successful? Decision point – Was the Work Order created successfully? If yes, go to Step 50. If no, go to Step 60.	S/4HANA	S/4HANA	
50	Create Customer Contact A Customer Contact is automatically created to log the successful creation of the Work Order for the customer. Once complete, go to Step 120.	S/4HANA BCT0	S/4HANA	
60	Create Case If the creation of the Work Order fails, an automatic BPEM case is created. Once complete, go to Step 120.	S/4HANA EMMAC1	S/4HANA	
70	Call Routed to Agent	N/A	Field Work Order Processing – CCC/BCD	
80	Manage Business Partner	Hybris C4C BP	Field Work Order Processing – CCC/BCD	
90	Premise Required? Decision point – Is the creation premise required to create the Work Order?	Hybris C4C	Field Work Order Processing – CCC/BCD	
100	Create Work Order	Hybris C4C IW31	Field Work Order Processing – CCC/BCD	

Step	Description of Step	T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
110	Create Activity An activity will be manually created to log the successful creation of the Work Order for the customer. Once complete, go to Step 120.	Hybris C4C	Field Work Order Processing – CCC/BCD	
120	Manage Field Orders Orders will be fulfilled by DF-WM-02 .	S/4HANA	Field Work Order Processing	
130	End The process ends.	N/A	N/A	

2.7 3rd Party and SCE Legacy System Integrations

3 rd Party or SCE Legacy System Integration	Description of Integration and Required Change to 3 rd Party or SCE Legacy System	Estimated Complexity of 3 rd Party or SCE Legacy system change
Click		
OMS		

2.8 Data Elements

2.8.1 Service Order Data Elements

Please refer to the BPD DF-WM-02 - Handle Field Orders - Manage Field Orders for detail data elements.

2.8.2 Source Systems and Data Volumes

Currently it is estimated around almost 50,000 created in the last 6 months.

2.9 Cloud for Customer (C4C) Integration

2.9.1 Processes

Work Orders will be viewable from C4C Front end system, Orders that are linked to a premise and/or customer are viewable with historical dates, status (open vs. completed, etc.). New Work Order can be created, changed or cancelled via C4C system.

2.9.2 C4C Views

Screen will be available in C4C system where agent can view Work orders. Also option will be available for front office agent to create, change and cancel work order.

2.9.3 Activity

A new activity can be created when a new work order created or changed in the system.

2.10 Exception Management Requirements

This section outlines a high level description of each BPEM Case or other user workflow items (such as “Tickets” used by Front Office) used to route work requests between business areas. The companion document for this section which outlines the detail of each case is documented in APSE on the [Exception Case Template General Document #8](#).

User Exception Scenario / Description	User Group	Initiated by	Work Distribution
N/A			

2.11 Process Risks and Mitigation

Risk Statement	Mitigation
N/A	N/A

2.12 Key Performance Indicators

The following have been identified as KPI’s at Southern California Edison, along with the means of measuring each KPI and the current measurement statistic for each KPI:

Measured KPI	Description	Measurement Method	Current Statistic
N/A			

2.12.1 Key Metric Definition

The following have been identified as key metrics at Southern California Edison that require business rule for calculation:

Key Metric	Description	Business Rule	Reference
N/A	N/A	N/A	N/A

3 Controls, Regulatory Rules, and Security

Requirement ID (from RTM)	"As-Is" Control Requirement	"To Be" Risk Statement	"To-Be" Control Activity	Control Automation (manual, IT dependent- manual, automated)	Relevant technical identifier or RICEFW reference
N/A	N/A	N/A	N/A	N/A	N/A

3.1 Security Considerations

Role	Transaction Code (T-Code)	T-Code Description	Functional Restriction	Organizational Restriction
Field Work Order Processing – All Users	C4C (Workcenter: Call Center)	Display Work Order	Display	
Field Work Order Processing – CCC/BCD/RSO	IW31	Create Work Order	Create	
Field Work Order Processing – CCC/BCD/RSO	C4C (Workcenter: Call Center)	Create Work Order	Create	
Field Work Order Processing – CCC/BCD/RSO	C4C (Workcenter: Call Center)	Change Work Order	Change/Cancel	
Field Work Order Processing – CCC/BCD/RSO	C4C (Workcenter: Call Center)	Display Work Order	Display	
Field Work Order Processing – CCC/BCD/RSO	IW31	Create Work Order	Create	
Field Work Order Processing – CCC/BCD/RSO	IW32	Change Work Order	Change/Cancel	
Field Work Order Processing – CCC/BCD/RSO	IW33	Display Work Order	Display	
Field Work Order Processing – Credit/Collections/Billing/CCC/BCD	IW33	Display Work Order	Display	

3.2 Custom Program/Transaction Codes

Custom Table	Authorization Grp Assigned	Custom T-code for the Table	Table Access (display/update)	RICEFW Reference ID
N/A	N/A	N/A	N/A	N/A

3.3 Custom Table/Transaction Codes

Custom Table	Authorization Grp Assigned	Custom T-code for the Table	Table Access (display/update)	RICEFW Reference ID
N/A	N/A	N/A	N/A	N/A

3.4 Data Criticality and Cybersecurity Baseline Risks

Specify Data Classification /Special Handling or Regulatory Requirement <i>What is the classification of Data? Does it trigger any special regulations?</i> <i>Please select the classification / Sensitivity of the Data. This will help us Assess the Risk associated and remediation measures if any.</i>	<Select all that apply> ☐ PII - Personal Identifiable Information / Privacy Law ☐ CEII - Critical Energy Infrastructure Information ☐ FERC SOC (Standards of Conduct) ☐ NERC CIP Confidential ☐ Sarbanes-Oxley (SOX) financial reporting data ☐ SCE utility data accessible by affiliate companies (CPUC) ☐ PCI - Payment Card Industry Compliance ☐ Customer Information Service Request (CISR, CISR-DRP) ☐ Other <u><Please Specify></u> ☒ Internal ☐ Public ☒ Confidential
Please identify which business Group/Role will require access to the data and/or solution: <i>Please select the list of Busines Groups will have access to this Data / Process. This will help us Determine the Impact and Risk Spread. Will ensure efficient Risk mitigation measures.</i>	<Select all that apply> ☐ SCE Customer, Res./Comm. Customer; ☒ SCE OU, Business Function Organization Name: CS, T&D ☒ Vendor: ELMS ☐ External business partner(s) & SaaS providers ☒ External on-shore contractors ☒ External off-shore contractors ☐ SCE Affiliate Company Employees ☒ SCE Employees, specify if they have Marketing Function responsibilities (e.g., ES&M) CCC, BCD ☒ SCE contract workers - ELMS ☐ Regulatory Agencies ☐ Others <u><Please Specify></u>
How many of the SCE Customer, Employee, Vendors will use or affected by the Business Process?	1000+

<i>To identify the Impact if services are adversely affected.</i>	
How much Information/data would be processed, received or transmitted by the Business Process? <i>The data volume that will be utilized by this process. Helps us in determining the Impact and criticality</i>	
Number of business transactions in average? <i>These are the number of transactions process by the application on average. The number of transactions help in understanding criticality and Impact.</i>	Today's Basic Work Request that are sent to field (does not include Trouble Orders) 2017 - to Field - 180,889 2016 - to Field - 168,718 2015 - to Field - 204,991 ELMS Work Orders- 683,889 for years 2004-2018, approx 48,849 per year set by field
Additional Security Consideration:	Any Cybersecurity impact in addition to SAP system security: • Internet Facing Web Application? No • Internet Facing Application Programming Interface? No • 3rd party Federated Authentication & Authorization, Single Sign On? No

4 Technical Considerations

4.1 RICEF-W Assessment

4.1.1 Reports

Report Name & Description (List Requirements met by this Report)	Complexity	Reference ID
N/A	N/A	N/A

4.1.2 Interfaces

Interface Name & Description (List Requirements met by this Interface)	Complexity	Reference ID
N/A	N/A	N/A

4.1.3 Conversion

Conversion Name & Description (List Requirements met by this Conversion)	Complexity	Reference ID
N/A		

4.1.4 Extension/Enhancements

Extension Name & Description (List Requirements met by this Extension)	Complexity	Reference ID
Ability for user (creator or other system users) to subscribe to receive status alert for specific request	M	E-CIM-CS-05 Work Order Notification
Work Order Creation with all required parameters.	M	E-CIM-CS-05 Order Creation
Ability for the user to issue a request using the following: BP master data, Technical Master data, Free Form Notes, Category / Action Required / Goal (for sales Task replacement) or Type and Sub type	M	E-CIM-CS-05 Request Creation
The users shall be able to (global) search for transaction using multiple selection criteria which includes (removed) types and sub-types (Action Requested and Goal/Reason) and Based on parent work order, system will automatically create individual child work orders and route to appropriate group	M	E-CIM-CS-05 Request Search

4.1.5 Forms

Form Name & Description (List Requirements met by this Form)	Complexity	Reference ID
N/A	N/A	N/A

4.1.6 Workflows

Workflow Name & Description (List Requirements met by this Workflow)	Complexity	Reference ID
N/A	N/A	N/A

4.2 IDEX Considerations

N/A

4.3 Production System Cut-over Considerations

- Active or future work order will be converted into the SAP system
- Employees will be able to view historical work orders via CSSWS/CWO for a minimum of 12 months and be able to query for a report after the system is disabled.

4.4 Other Considerations

N/A

5 Change Management

5.1 Change Impacts

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
Business Customer Division (BCD) will have the opportunity to issue the work order directly to the field instead of an email or front office task (ex voltage, power quality, cross meter, etc.). Currently BCD will send a tasks or emails to COE. The capability will require more training for process but will reduce handoffs.	Benefit/Improvement; Process (457)	CS-CSOD-CCC All. CS-BCD-All	Low	Low
Front office will have the ability to schedule specific appointment times for customers for service orders as well as field inspections which provides the customer flexibility and increases customer satisfaction Customer Impact: Provides customers flexibility and customer satisfaction	Benefit/Improvement; Decision Making (460)	CS-BCD-All; CS-CSOD-CCC All; Ext-Customers-All; T&D-All	Medium	Medium
The system will be able to issue an automated "add lock ring" request after a certain amount of days once a lock ring removal request has been completed which will minimize the amount of call backs to the contact center. Currently customers need to call back to request to add the lock ring once the electrical work has been completed or agents will need to tab on their calendars to add the lock ring back on the meter.	Benefit/Improvement; Process (461)	CS-CSOD-CCC All, CS-BCD-All	Low	Low

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
Field service orders do not have a separate access arrangements field. Comments field within order is currently limited to 40 characters. The user would need to go into the back office if longer text is required. Agents will need to use the comments field to be concise on the request and will need to focus the access arrangements at the premise level. Currently there is an access field in the field basic work requests that download into the field tools to be viewed (eg locked gates, dogs etc). Users will need to be trained to how to update the access arrangements.	Decision Making;Process (836)	CS-BCD-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs	Medium	Medium
Front office users will have the ability to create a service order for a non-sce customer using the FOP launcher including non-standard addresses. Currently non-sce customers can call and request service orders which can be linked to the closest installed service. This is a change in process that will require users to access the back office UI.	Culture Shift;Process (837)	CS-BCD-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs	Low	Low
Changes to technical master data will make changes to all appropriate levels automatically. Currently authorized users can manually change the Installed service and site levels but the service account is not updated. This causes the service address and the other levels to not match the premise level.	Benefit/Improvement;Process	CS-BCD-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs;CS-CSOD-RSO All	Low	Low

Below is a description of each:

Change Element Short Description: Describes the ‘As-Is’ scenario and what is changing.

Change Impact Category: Determined by asking the questions below.

1. **Staffing Levels** - Is the work effort increasing or decreasing (staffing levels), and how (e.g., task automation, new process/tasks)?

2. **Job Roles** - Will the responsibilities, authority, or required skills for a job or job roles change? Will people be reporting to new leaders? Will managers' responsibilities change?
 3. **Compensation / Performance** - Will the changes have such an impact on roles of an individual that merits a change in how they are compensated or how their performance will be evaluated? What behaviors are rewarded and recognized?
 4. **Organization Structure** - Is the ownership of the Process or Data changing (think departments, jobs, or roles, not individuals)?
 5. **Decision Making** - Will any of the changes impact how decisions are made, by or between, business units / departments? Who makes them, how they determine the criteria for making a change?
 6. **Culture Shift** - Will organizational value systems impact the way change happens or be impacted by the change? What are the underlying values of the organization that will directly influence the way change will be accepted and how much work will ultimately be required to ensure successful outcomes for the business?
 7. **Location / Facilities** - Will the change impact the physical location where they process is being implemented or require changes to the facilities to enable the process to be effective?
 8. **Process** - Any change to the current process (e.g. manual process becomes automatic, or vice versa)? Are there new processes being introduced or an old process being decommissioned? Is there a requirement to change an existing business rule or business policy? Will new business rule or policy needs to be introduced and enforced or complied to?
 9. **Regulatory/ Legal** - Are there any legal / regulatory requirements that might be impacted
 10. **Technology** - Are other applications impacted (e.g., going away, becoming read-only), or are there new requirements (e.g., new SW / HW, new version)
 11. **Benefit/Improvement** - Will there be an improvement or benefit to any of the stakeholders because of this change. (Be sure to include any pain points that are being alleviated)
 12. **Other** - Are there emotional implications stemming from any of these changes (think of buy-in issues or fears)
- Audience Group(s):** These are the affected departments, divisions etc...
- Impact:** The following were used to determine the impact:

Level	
HIGH 	BUSINESS OUTCOMES: Drives change to cost, efficiency, customers. JOB: More than 40% of individual's job will change and / or additional skills are required with significant changes to Job Documents. Examples include: - Shift from manually intensive role to using new tools and systems - Employee is going to be redeployed - High level of "technical or functional" changes needed for a particular group - Significant changes to culture & attitudes to get job done - Big shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) RELATIONSHIPS: High level of "relational" changes for a particular group (e.g. reporting relationships, effective working relationships with customers, teams, and stakeholders etc.) POLICIES/PROCEDURES: Processes, procedures, and/ or policies are not in place today and need to be defined. REGULATORY REQUIREMENT: drive changes to the way we comply

Level	
MEDIUM 	• BUSINESS OUTCOMES: Limited change to cost, efficiency, customers. • JOB: 20-40% of individual's job will change and / or additional skills are required with significant changes to Job Documents, Examples include: • Employee job title or function is modified, resulting in minor to moderate changes to roles and responsibilities. • There will be some changes to the culture & attitudes • Shift from moderate technology use to using new tools and systems • Moderate level of "technical or functional" changes needed for a particular group • Moderate shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) • RELATIONSHIPS: Moderate level of "relational" changes for a particular group (e.g. reporting relationships) • POLICIES/PROCEDURES: Processes, procedures, and/or policies are inconsistent today and not defined or documented in many cases • REGULATORY REQUIREMENT: No impact.
Level	
LOW 	• BUSINESS OUTCOMES: No impact. • JOB: Less than 20% of individual's job will change and / or additional skills are required with significant changes to Job Documents. Examples include: • Employee job title or function is slightly modified with few, if any, changes to roles and responsibilities • There will be little change to the culture & attitudes • Shift from using some technology to new systems & tools • Light level of "technical or functional" changes needed for a particular group • Small shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) • RELATIONSHIPS: No impact or light level of "relational" changes for a particular group (e.g. reporting relationships) • POLICIES/PROCEDURES: Processes, procedures, and/or policies are consistent, but not well documented, and will need to be documented/ Other minor changes to be made. • REGULATORY REQUIREMENT: No impact.

Business Effort: Initial assessment of the impact to the business High, Medium, Low

6 Requirements



Design Phase RTM
CIM-CS-05-Handle Re



Proc Owner
Approval_CIM_CS_05_